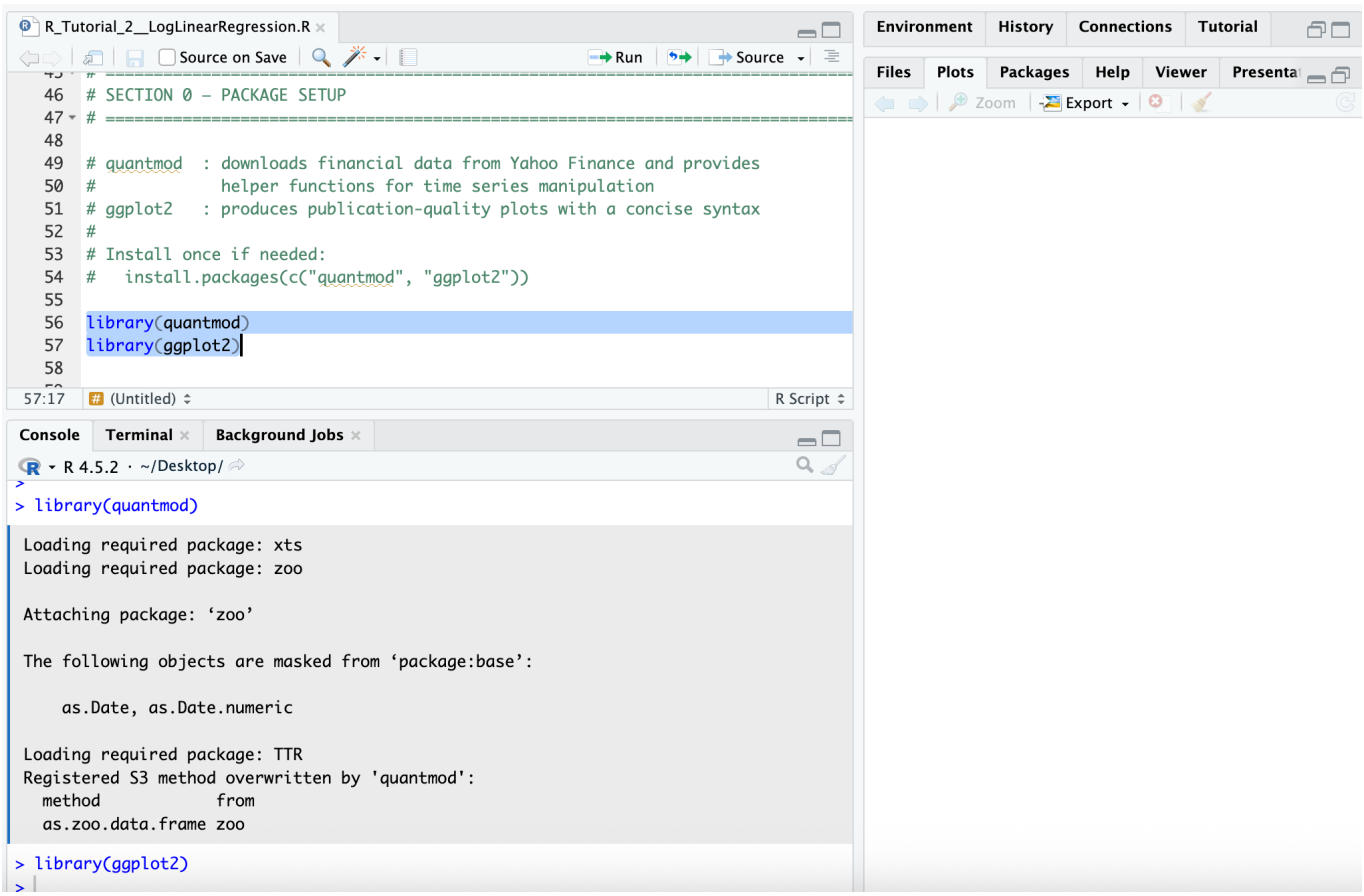
Financial Econometrics, R Tutorial 2

Run a log-linear regression of a financial time series on a time trend

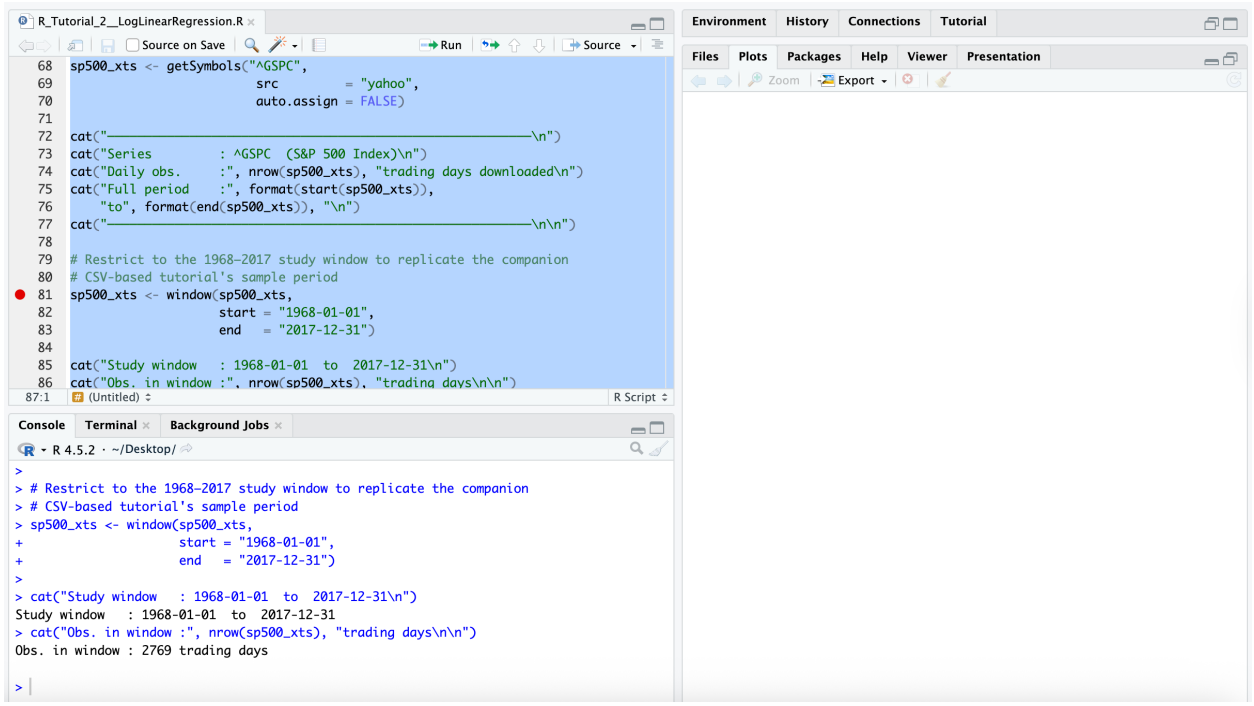
Welcome to tutorial 2. In this tutorial, you will learn how to run basic R functions and estimate regression models in RStudio. The instructions apply to both Windows and macOS successfully, although they may look slightly different.

Note: If you see  any screenshot, use it as a reference point to identify where to click on your R / similar window.

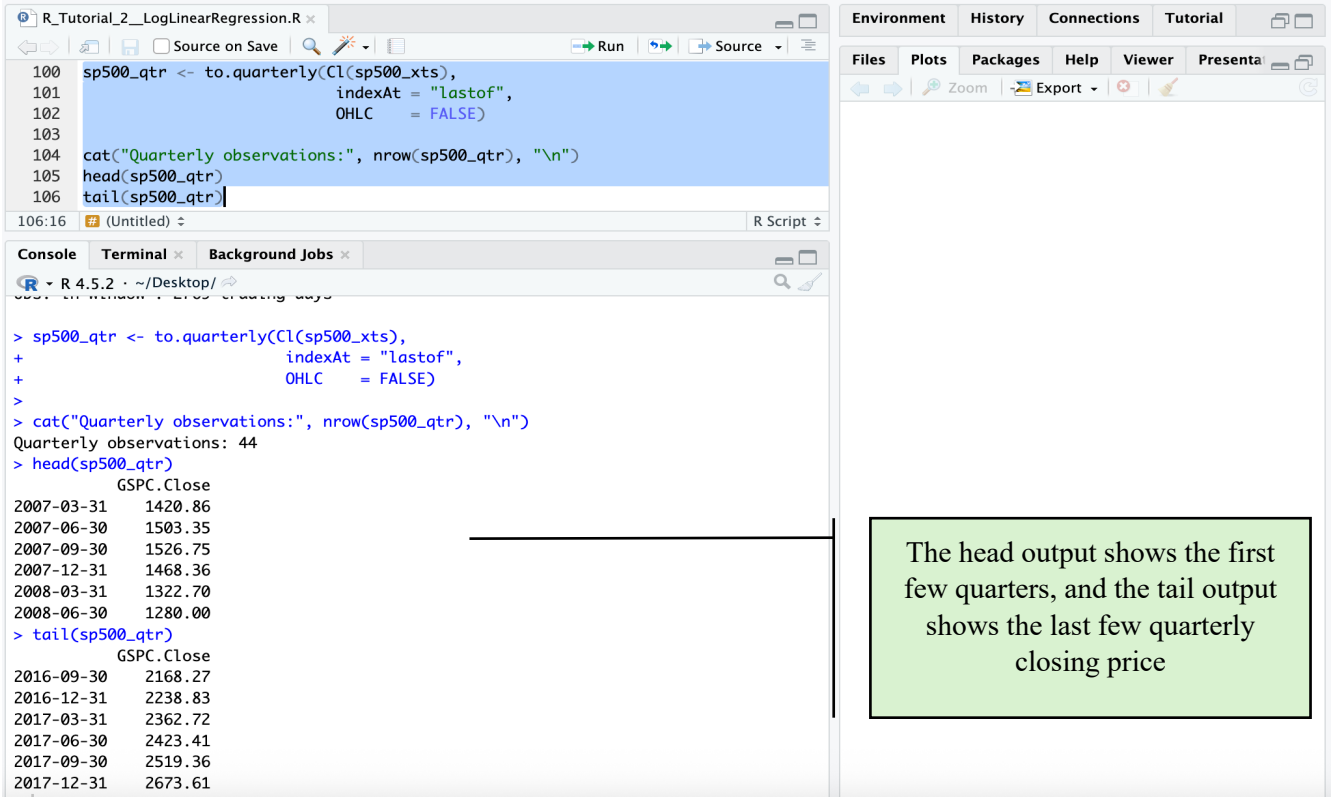
1. Open the R code file called R_Tutorial_2_LogLinearRegression.R in RStudio. First, you need to make sure that the quantmod and ggplot2 packages are installed before running other code. After opening the R code file, you should see the following:



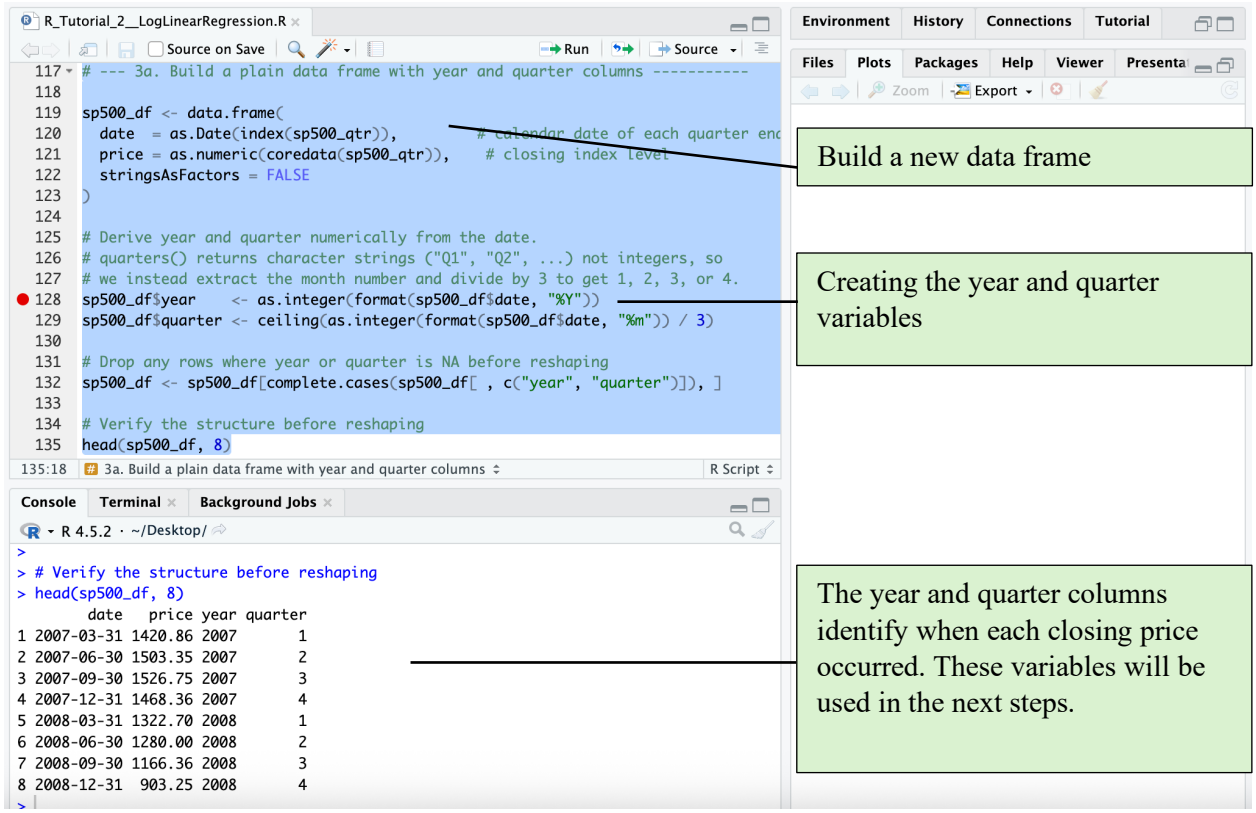
2. We downloaded daily S&P 500 Index data directly from Yahoo Finance and selected the period from 1968 to 2017. The selected daily data will be prepared for later steps of regression analysis. Run the blue screenshot, you will see:



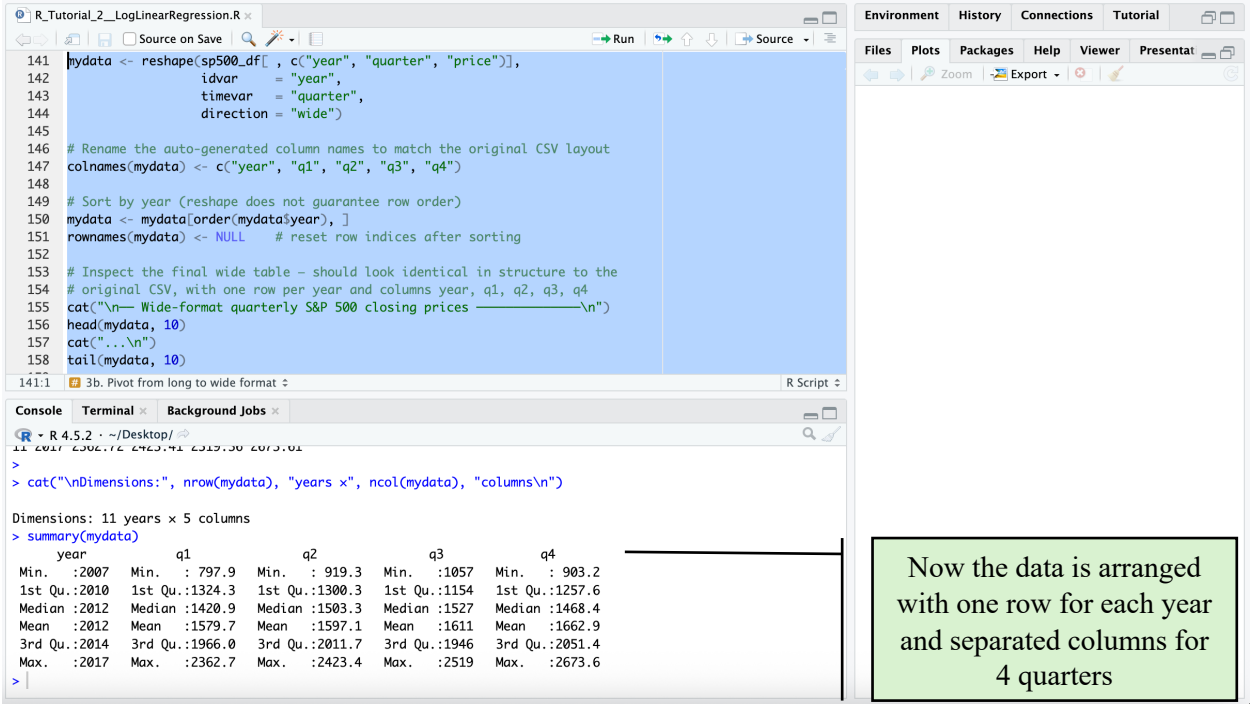
3. The data downloaded from Yahoo Finance is daily data, but we want to convert the daily S&P 500 closing prices into quarterly observations. The code uses the last available closing price of each quarter as the quarterly value. Running this code and you will see:



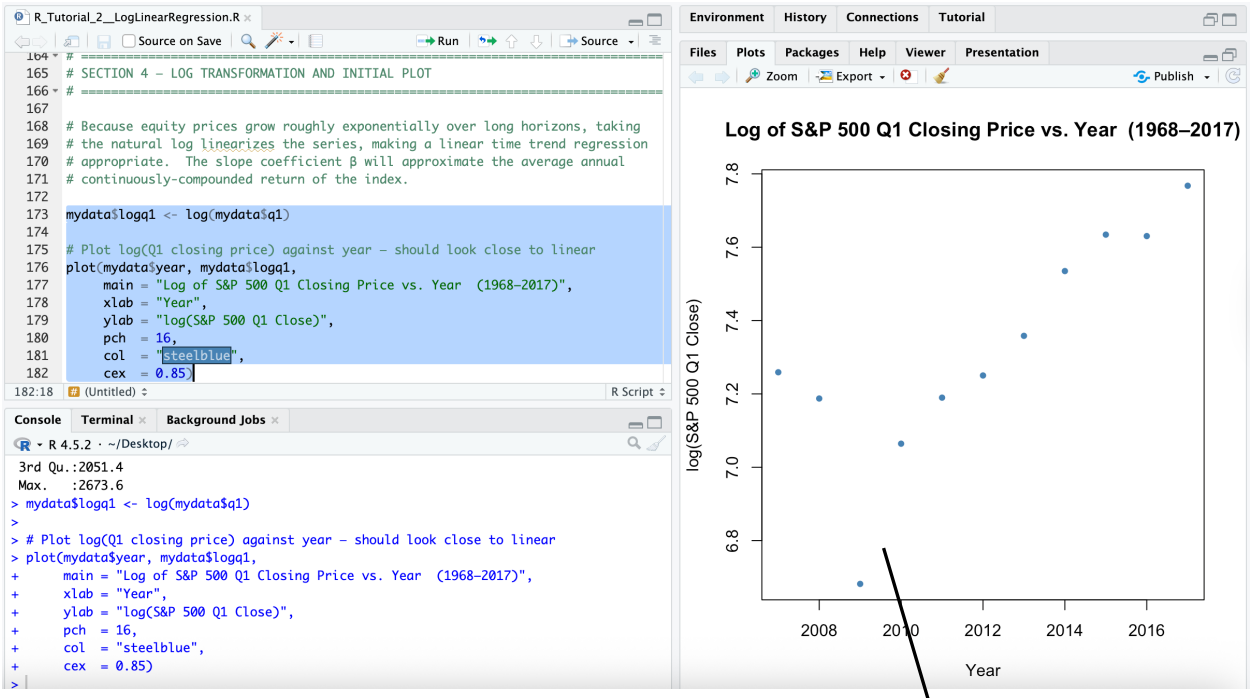
4. Before reshaping the data, we first convert the quarterly price series into a regular data frame. Each row shows its own quarter and contains its date and closing price. And after that, we create year and quarter variables from the date so that the data can be rearranged in the next step.



5. We want to reshape the quarterly data from the previous long format to a wide format for future regression. After reshaping, each row represents one year, and the quarterly closing prices are stored in separate q1, q2, q3, and q4 columns. After running the highlighted code, you will see:

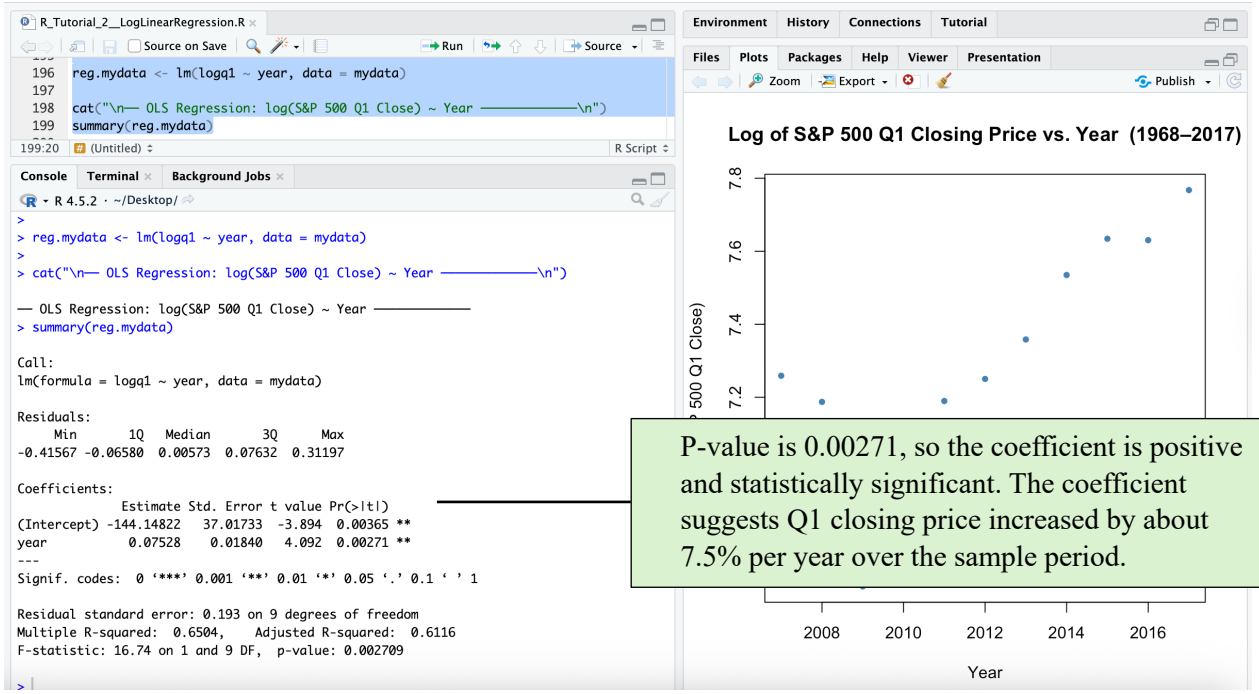


6. Before running the regression, we take the natural log of the Q1 closing prices as our data and plot them over the years. This helps us examine whether the long-run growth pattern is approximately linear before running the regression.

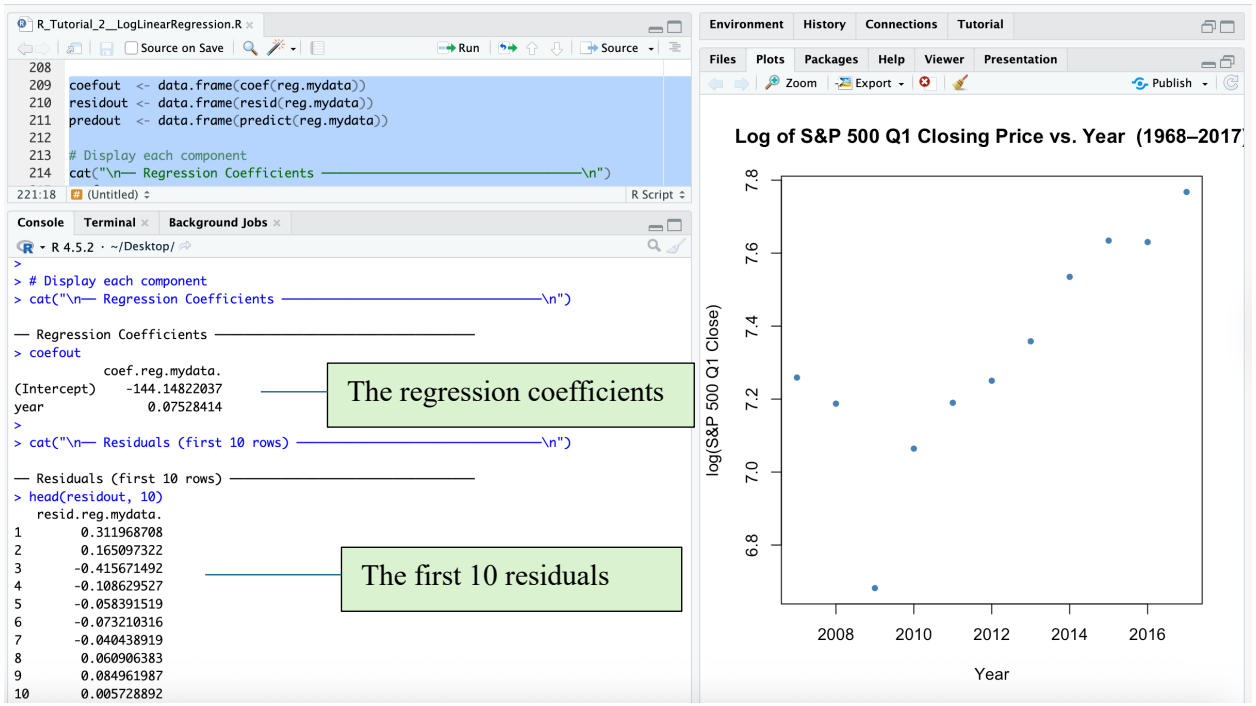


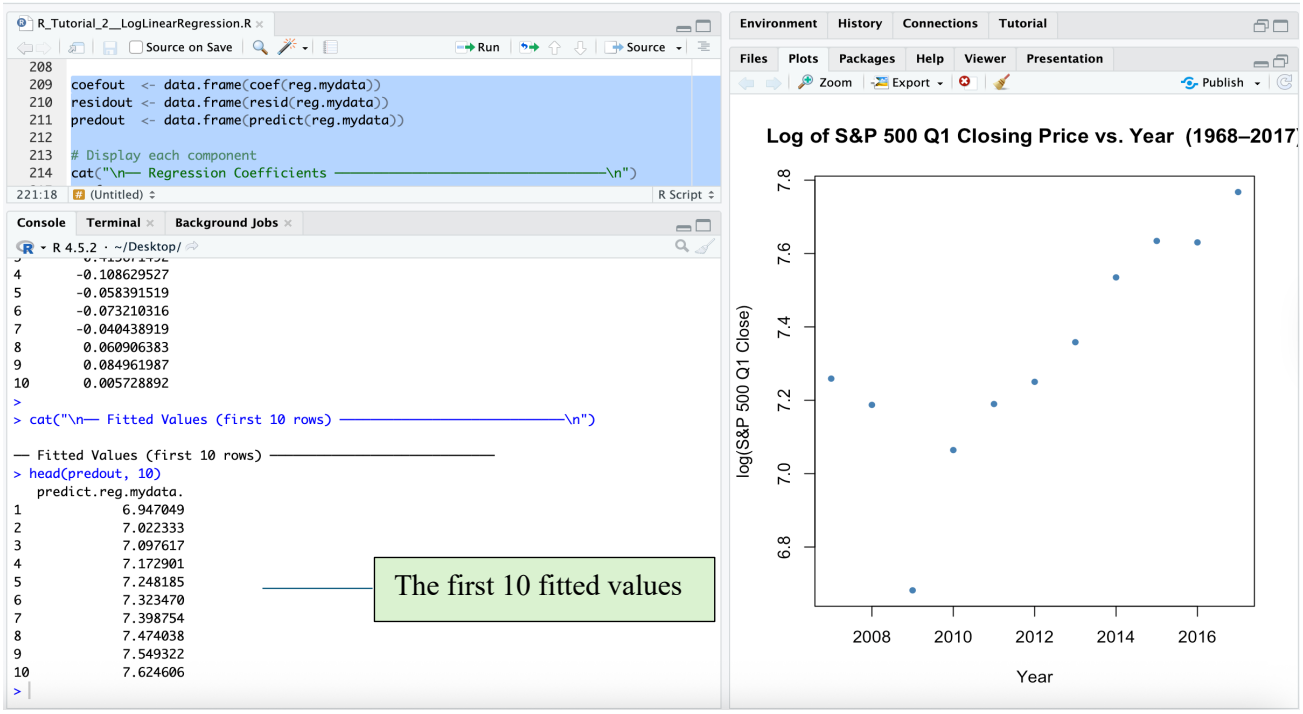
The plot shows the log Q1 closing prices over time. The points suggest an upward pattern, which makes the log-linear regression reasonable.

7. We can now run the regression. After running the highlighted code, you should see the regression output below:

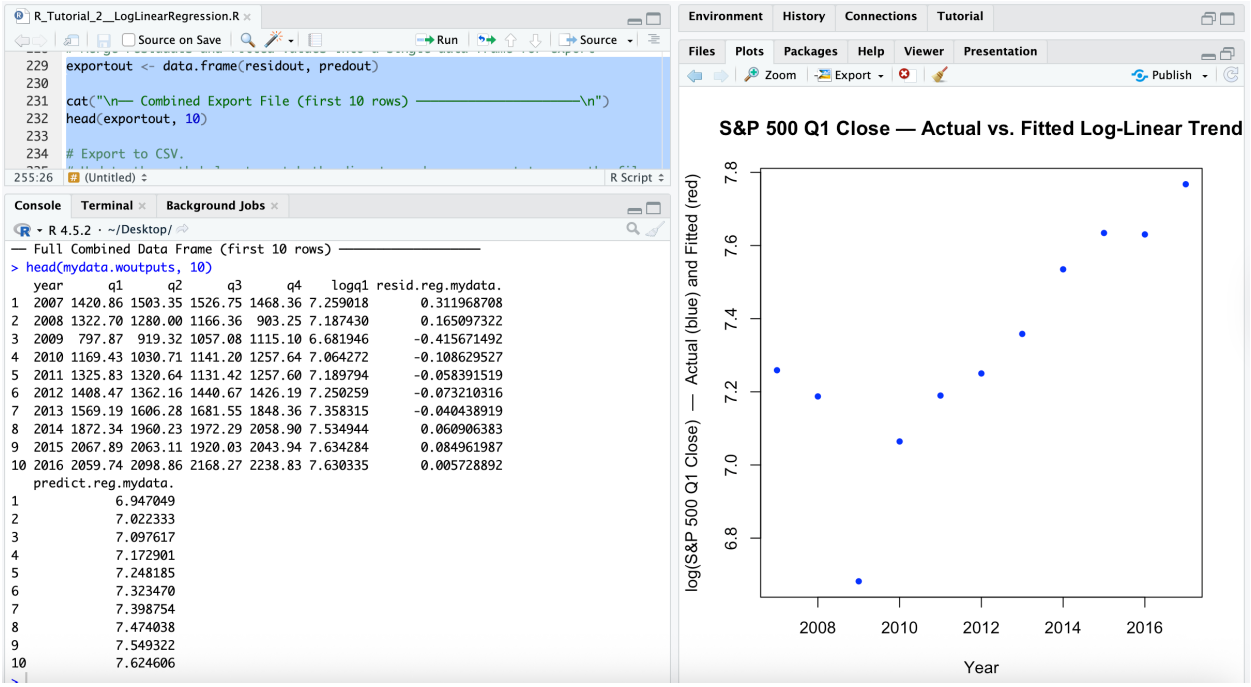


8. After running the log-linear regression model, we want to extract three important outputs for future comparison: the estimated coefficients, the residuals, and the fitted values. After running the highlighted code, you should see the following:

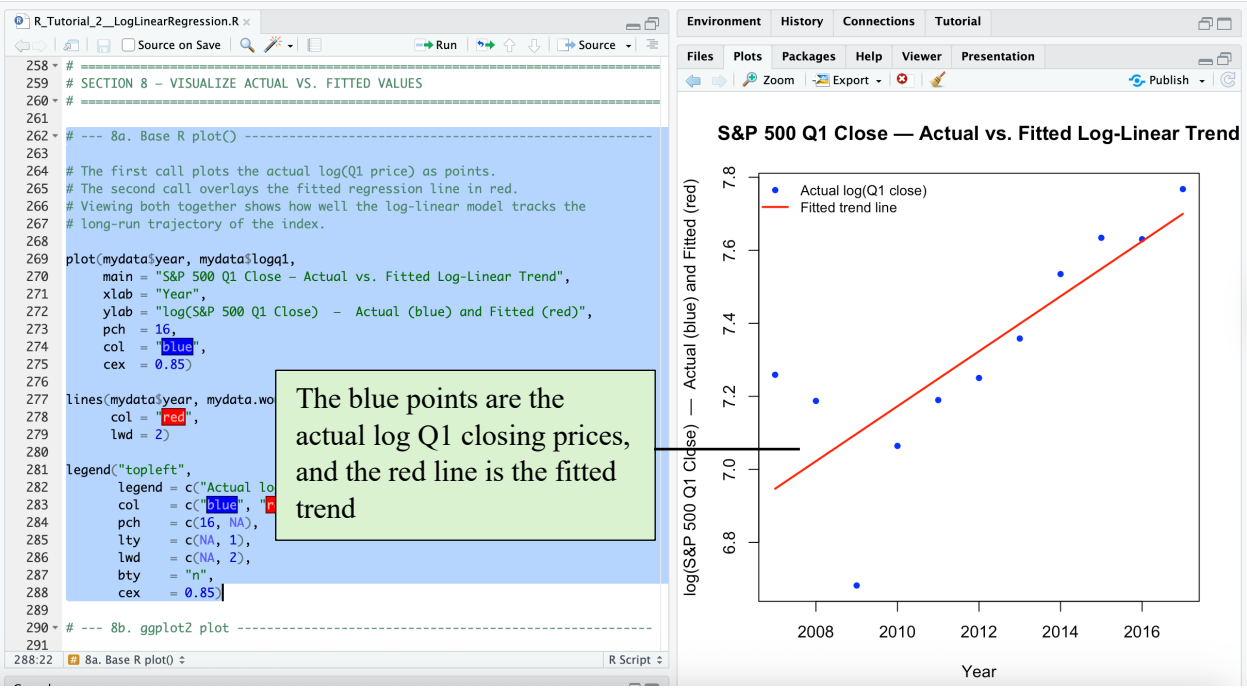




9. Now we can combine the regression residuals and fitted values into data frames and export them. We combine the original data with the regression outputs, which will be used for plotting, running section 7 and you will see:



10. We now plot the actual log-transformed Q1 closing prices and their fitted values at the same time. After running the blue highlighted code, you should see the following:



11. We want to put the actual log Q1 closing prices into a solid line. And run the following code, you will see the blue solid line represents the actual values of logq1, while the red dashed line represents the fitted log-linear trend, and the gap between them reflects the residuals:

